

Mathematical Methods for International Commerce

Week 13/2: Basic Matrix Operation (7.1)
Matrix Inversion (7.2)

Igor Vyshnevskyi, Ph.D.

Sogang University, Geppert School of International Studies

May 28, 2026

Agenda

1. Basic Matrix Operation (7.1)
2. Matrix Inversion (7.2)
3. Homework #2



Why Matrices Matter in Economics

- Matrix algebra allows economists to:
 - Solve systems of equations
 - Model economic input-output relationships
 - Optimize production & costs
 - Forecast using linear models
- Applications include:
 - Input-output models
 - Markov chains
 - Linear regression models
 - Economic equilibrium analysis



The seal of Gwangju University is a circular emblem. It features a central shield with a crown on top and the letters 'IHS' inside. The year '1960' is inscribed above the shield. The outer ring of the seal contains the university's name in Korean and English: '광주대학교' and 'GWANGJU UNIVERSITY'.

1. Basic Matrix Operation (7.1)

Example: Sales Table

Suppose that a firm produces three types of goods (G1, G2, G3) and sells them to two customers (C1 and C2). The matrix:

$$A = \begin{bmatrix} 7 & 3 & 4 \\ 1 & 5 & 6 \end{bmatrix}$$

represents monthly sales where:

- Rows: C1, C2 (customers)
- Columns: G1, G2, G3 (goods)

This format allows for compact storage and easy operations like summing totals or multiplying by price vectors.

Basic Matrix Terminology

- A matrix is a rectangular array of numbers:

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$$

- **Order:** number of rows $\times$ number of columns (dimensions)
- **Row vector:** 1 row, n columns
- **Column vector:** n rows, 1 column
- **Element:** a_{ij} is the element in row i , column j
- **Square matrix:** same number of rows and columns (e.g., 2×2)
- **Zero matrix:** all elements are zero

$$O = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

General Matrix Notation

A general matrix D of order 3×2 :

$$D = \begin{bmatrix} d_{11} & d_{12} \\ d_{21} & d_{22} \\ d_{31} & d_{32} \end{bmatrix}$$

A general matrix E of order 3×3 :

$$E = \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix}$$

Basic Matrix Operations

Transpose of a Matrix

- **Transpose:** A^T flips rows and columns

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \Rightarrow A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

- Rows become columns
- Used frequently in optimization and econometrics

Basic Matrix Operations (continued)

Matrix Addition & Subtraction

Two matrices of the same order:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}, \quad A - B = \begin{bmatrix} -4 & -4 \\ -4 & -4 \end{bmatrix}$$

Basic Matrix Operations (continued)

Scalar Multiplication

Multiply each element:

$$2A = 2 \cdot \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$$

Basic Matrix Operations (continued)

Matrix Multiplication

- Only defined if inner dimensions match: $A_{m \times n} \cdot B_{n \times p}$

Example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, B = \begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 \cdot 2 + 2 \cdot 1 & 1 \cdot 0 + 2 \cdot 5 \\ 3 \cdot 2 + 4 \cdot 1 & 3 \cdot 0 + 4 \cdot 5 \end{bmatrix} = \begin{bmatrix} 4 & 10 \\ 10 & 20 \end{bmatrix}$$

- Dimensions of the result: $m \times p$
- | Matrix multiplication is NOT element-by-element multiplication.

Basic Matrix Operations (continued)

Matrix Multiplication: Row × Column

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix}, \quad AB = \begin{bmatrix} 4 & ? \\ ? & ? \end{bmatrix}$$

The **first element** c_{11} comes from **Row 1 of A** × **Col 1 of B** :

$$c_{11} = 1 \cdot 2 + 2 \cdot 1 = 2 + 2 = 4$$

In general: $c_{ij} = \text{row}_i(A) \cdot \text{col}_j(B)$ — each entry is a **dot product**.

| Matrix multiplication is **not** element-by-element — it is row × column interaction.

Basic Matrix Operations (continued)

Matrix Multiplication: Dimension Check

Before multiplying, verify dimensions:

- **Inner dimensions must match:** $A_{m \times n} \cdot B_{n \times p}$ is valid
 - **Result dimensions:** $m \times p$ (outer dimensions)
-

Example:

$$A : 3 \times 5, \quad B : 5 \times 2, \quad C : 3 \times 4$$

✓ $A \cdot B$: inner is $5 = 5$ ✓ → result is 3×2

✗ $A \cdot C$: inner is $5 \neq 3$ ✗ → **not defined**

Basic Matrix Operations (continued)

General Matrix Multiplication

If A is $m \times s$ and B is $s \times n$, then AB is $m \times n$. Element c_{ij} is:

$$c_{ij} = \text{row}_i(A) \cdot \text{col}_j(B)$$

Let:

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 1 & 2 & 1 \\ 1 & 0 & 1 & 2 \\ 5 & 4 & 1 & 1 \end{bmatrix}$$

Check: A is 2×3 , B is $3 \times 4 \Rightarrow AB$ exists, size is 2×4

Basic Matrix Operations (continued)

Calculating Elements of AB

- $c_{11} = 2 \cdot 3 + 1 \cdot 1 + 0 \cdot 5 = 6 + 1 + 0 = 7$
- $c_{12} = 2 \cdot 1 + 1 \cdot 0 + 0 \cdot 4 = 2 + 0 + 0 = 2$
- $c_{13} = 2 \cdot 2 + 1 \cdot 1 + 0 \cdot 1 = 4 + 1 + 0 = 5$
- $c_{14} = 2 \cdot 1 + 1 \cdot 2 + 0 \cdot 1 = 2 + 2 + 0 = 4$

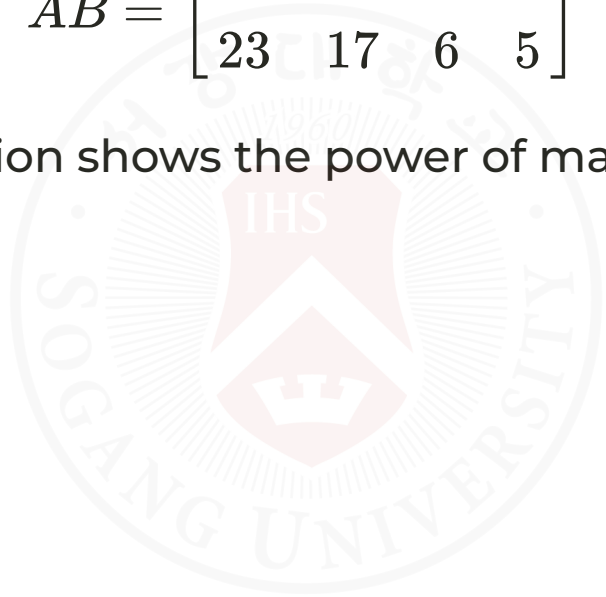
Continue for second row...

Basic Matrix Operations (continued)

Full Product AB

$$AB = \begin{bmatrix} 7 & 2 & 5 & 4 \\ 23 & 17 & 6 & 5 \end{bmatrix}$$

Step-by-step matrix multiplication shows the power of matrix algebra in summarizing economic relationships.



Properties of Matrix Operations

- A matrix is a rectangular array of numbers, organized into rows and columns.
- The dimensions of a matrix are given as $m \times n$, where m is the number of rows and n is the number of columns.
- Each element in the matrix is indexed by its row and column position, denoted as a_{ij} .

Provided that the indicated sums and products make sense,

- $A + B = B + A$
- $A - A = O$ where O is the zero matrix
- $A + 0 = A$
- $k(A + B) = kA + kB$
- $k(lA) = (kl)A$
- $A(B + C) = AB + AC$
- $(A + B)C = AC + BC$
- $A(BC) = (AB)C$

We also have the **non-property**:

- $AB \neq BA$

Matrix Representation of Systems

System of equations:

$$\begin{cases} 2x + 3y = 8 \\ 4x - y = 2 \end{cases} \Rightarrow AX = B$$

Where:

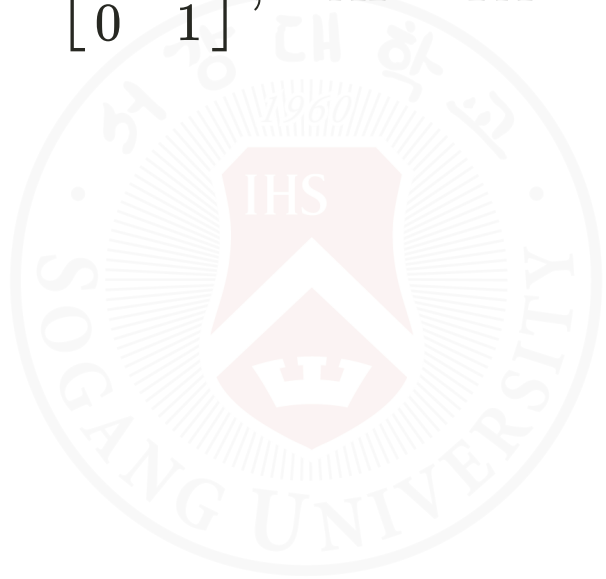
$$A = \begin{bmatrix} 2 & 3 \\ 4 & -1 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} 8 \\ 2 \end{bmatrix}$$

- A : coefficient matrix
- X : vector of unknowns
- B : constants vector

Identity Matrix

- Identity matrix I acts like 1 in multiplication:

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad AI = IA = A$$



2. Matrix Inversion (7.2)



Matrix Inversion (2x2 Case)

Given:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

- The matrix A^{-1} is called the inverse of A , and it plays a role similar to the reciprocal of a number in arithmetic.
- Although the formula for A^{-1} may appear complex, its construction for a 2×2 matrix is straightforward and systematic.

If $\det(A) = ad - bc \neq 0$, the inverse is:

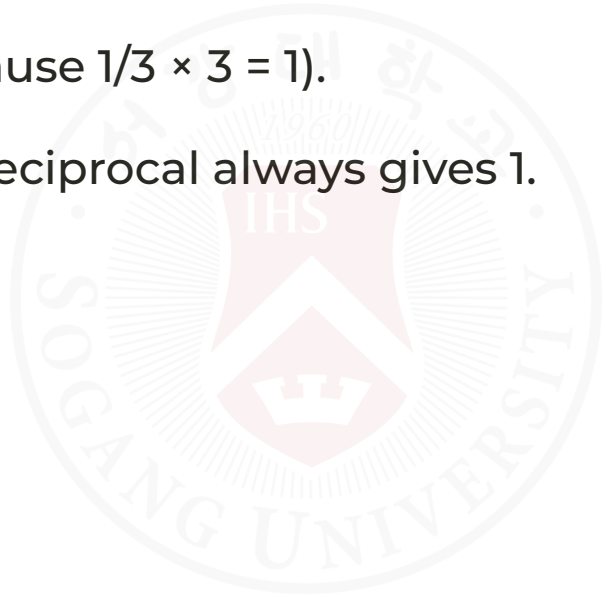
$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Note: $\det(A)$ is the determinant of A . If $\det(A) = 0$, the matrix is singular and cannot be inverted.

Nota bene

For any nonzero number x , its reciprocal is $1/x$.

- The reciprocal of 5 is $1/5$.
- The reciprocal of $1/3$ is 3 (because $1/3 \times 3 = 1$).
- Multiplying a number by its reciprocal always gives 1.



Solving Equations Using Inverses

From $AX = B$, multiply both sides by A^{-1} :

$$X = A^{-1}B$$

- This allows us to find the solution vector X directly.
- If A is invertible, we can solve systems of equations efficiently.

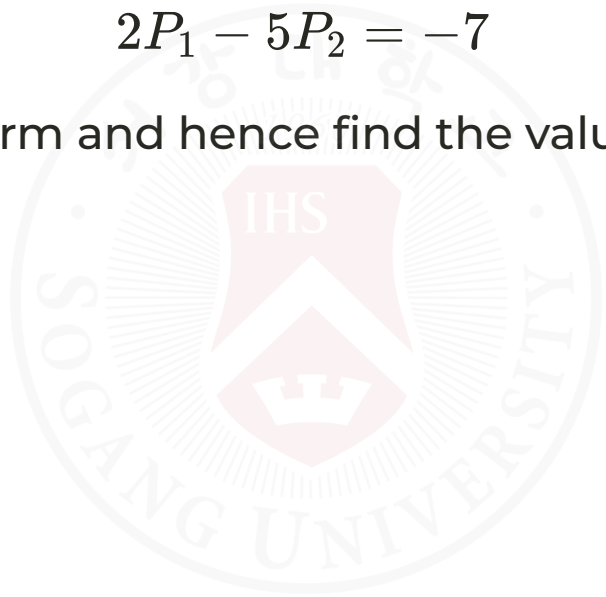
Example: Solving for Equilibrium Prices

We are given a system of equations:

$$-4P_1 + P_2 = -13$$

$$2P_1 - 5P_2 = -7$$

Express this system in matrix form and hence find the values of P_1 and P_2 .



Step 1: Express in Matrix Form

Write the system as:

$$\begin{bmatrix} -4 & 1 \\ 2 & -5 \end{bmatrix} \begin{bmatrix} P_1 \\ P_2 \end{bmatrix} = \begin{bmatrix} -13 \\ -7 \end{bmatrix}$$

Let:

$$A = \begin{bmatrix} -4 & 1 \\ 2 & -5 \end{bmatrix}$$

$$x = \begin{bmatrix} P_1 \\ P_2 \end{bmatrix}$$

$$b = \begin{bmatrix} -13 \\ -7 \end{bmatrix}$$

So the system becomes

$$Ax = b$$

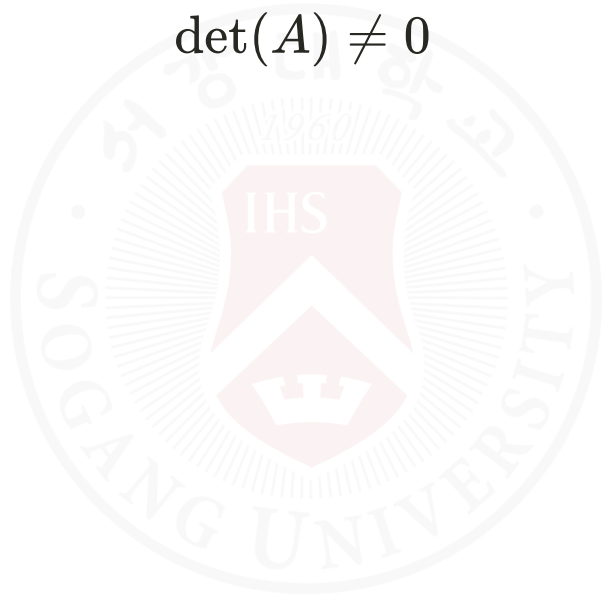
Step 2: Find the Determinant of A

$$\det(A) = (-4)(-5) - (1)(2) = 20 - 2 = 18$$

Since

$$\det(A) \neq 0$$

, the matrix is invertible.



Step 3: Find the Inverse of A

Using the formula for a 2x2 inverse:

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} -5 & -1 \\ -2 & -4 \end{bmatrix} = \frac{1}{18} \begin{bmatrix} -5 & -1 \\ -2 & -4 \end{bmatrix}$$

Step 4: Solve for x

Multiply

$$A^{-1}$$

by

$$b$$

:

$$\begin{bmatrix} P_1 \\ P_2 \end{bmatrix} = A^{-1}b = \frac{1}{18} \begin{bmatrix} -5 & -1 \\ -2 & -4 \end{bmatrix} \begin{bmatrix} -13 \\ -7 \end{bmatrix}$$

Step 5: Matrix Multiplication

Compute:

$$P_1 = \frac{1}{18}((-5)(-13) + (-1)(-7)) = \frac{1}{18}(65 + 7) = \frac{72}{18} = 4$$
$$P_2 = \frac{1}{18}((-2)(-13) + (-4)(-7)) = \frac{1}{18}(26 + 28) = \frac{54}{18} = 3$$

Final Answer

$$P_1 = 4$$

$$P_2 = 3$$

These are the equilibrium prices.



3x3 Matrices, Determinants, and Cofactors

The concepts of determinant, inverse and identity matrices apply to 3x3 matrices as well.

The identity matrix:

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Check that for any matrix A :

$$AI = A, \quad IA = A$$

To compute an inverse, we need **cofactors**:

- The cofactor A_{ij} equals the determinant of the corresponding minor multiplied by $(-1)^{i+j}$

3x3 Matrices, Determinants, and Cofactors (cont)

Example: For matrix A , to find A_{23} , delete row 2 and column 3 to get:

$$\text{Minor of } A_{23} = \begin{bmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{bmatrix}$$

Cofactor:

$$A_{23} = (-1)^{2+3} \cdot (a_{11}a_{32} - a_{12}a_{31}) = -a_{11}a_{32} + a_{12}a_{31}$$

This sign pattern follows the "checkerboard" rule:

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

3x3 Matrices, Determinants, and Cofactors (cont)

3x3 Determinants and Inverses

We are now in a position to describe how to calculate the determinant and inverse of a 3x3 matrix.

To compute $\det(A)$:

- Multiply elements in any row/column by their cofactors
 - The sum gives the determinant
 - Same result regardless of row/column used — useful for checking!
-

If expanding along the first row:

$$\det(A) = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}$$

Or down the second column:

$$\det(A) = a_{12}A_{12} + a_{22}A_{22} + a_{32}A_{32}$$

This flexibility allows for easier calculations in practice.

3x3 Matrices, Determinants, and Cofactors (cont)

3x3 Inverse Matrix Structure

The inverse of the 3×3 matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

is computed using cofactors and determinant.

1. Form the matrix of cofactors (the adjugate matrix):

$$\text{cof}(A) = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}$$

2. Take (1) transpose to get the adjoint matrix:

$$\text{adj}(A) = \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix}$$

3x3 Matrices, Determinants, and Cofactors (cont)

3x3 Inverse Matrix Structure (cont)

1. Multiply by $\frac{1}{\det(A)}$:

$$A^{-1} = \frac{1}{\det(A)} \cdot \text{adj}(A)$$

! If $\det(A) = 0$, the matrix is **singular** and the inverse **does not exist**.

Advice: Check your result by confirming that:

$$A^{-1}A = I \quad \text{and} \quad AA^{-1} = I$$

3x3 Matrices, Determinants, and Cofactors (cont)

Example: Cofactor Computation

Given:

$$A = \begin{bmatrix} 2 & 4 & 1 \\ 4 & 3 & 7 \\ 2 & 1 & 3 \end{bmatrix}$$

To find cofactor A_{11} : delete row 1 and column 1:

$$A_{11} = (+1) \begin{vmatrix} 3 & 7 \\ 1 & 3 \end{vmatrix} = 3(3) - 7(1) = 9 - 7 = 2$$

Similarly, all cofactors:

$$A_{11} = 2, \quad A_{12} = 2, \quad A_{13} = -2$$

$$A_{21} = -11, \quad A_{22} = 4, \quad A_{23} = 6$$

$$A_{31} = 25, \quad A_{32} = -10, \quad A_{33} = -10$$

3x3 Matrices, Determinants, and Cofactors (cont)

Example: Cofactor Computation (cont)

$$\text{adjugate}(A) = \begin{bmatrix} 2 & 2 & -2 \\ -11 & 4 & 6 \\ 25 & -10 & -10 \end{bmatrix}, \quad \text{adjoint}(A) = \text{adjugate}(A)^T = \begin{bmatrix} 2 & -11 & 25 \\ 2 & 4 & -10 \\ -2 & 6 & -10 \end{bmatrix}$$

3x3 Matrices, Determinants, and Cofactors (cont)

Example: Inverse of a 3x3 Matrix (cont)

Given $\det(A) = 10$, we compute:

$$A^{-1} = \frac{1}{10} \cdot \begin{bmatrix} 2 & -11 & 25 \\ 2 & 4 & -10 \\ -2 & 6 & -10 \end{bmatrix}$$

Verification:

$$A^{-1}A = I, \quad AA^{-1} = I \quad \checkmark$$

Practice Problems

- Given the matrix:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

- Find A^T (transpose).
 - Calculate $2A$ (scalar multiplication).
 - Add A to itself.
- Multiply:

$$\begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix}$$

- Find the inverse of:

$$\begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$$

- Solve for x and y :

$$\begin{cases} 3x + 2y = 10 \\ 4x - y = 5 \end{cases}$$

Practice Problems (continued)

- We are given a system of equations:

$$9P_1 + P_2 = 43$$

$$2P_1 + 7P_2 = 57$$

Express this system in matrix form and hence find the values of P_1 and P_2 .

- Determine whether the inverse exists for the matrix:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Brief Solutions (Practice)

- For $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$:

- $A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$

- $2A = \begin{bmatrix} 2 & 4 & 6 \\ 8 & 10 & 12 \end{bmatrix}$

- $A + A = \begin{bmatrix} 2 & 4 & 6 \\ 8 & 10 & 12 \end{bmatrix}$

- $\begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 2 & 5 \\ 3 & 20 \end{bmatrix}$

- $\begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}^{-1} = \frac{1}{10} \begin{bmatrix} 4 & -2 \\ -1 & 3 \end{bmatrix}$

- System $\begin{cases} 3x + 2y = 10 \\ 4x - y = 5 \end{cases}$ gives:

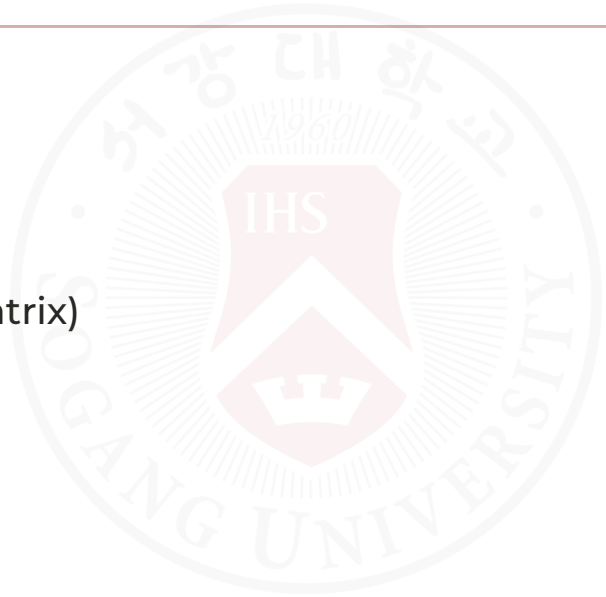
- $x = \frac{20}{11}, y = \frac{25}{11}$



Brief Solutions (continued)

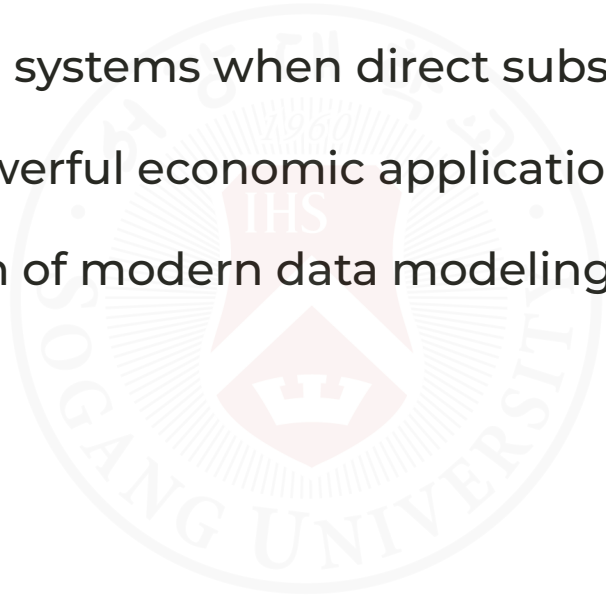
- System $\begin{cases} 9P_1 + P_2 = 43 \\ 2P_1 + 7P_2 = 57 \end{cases}$ gives:
 - $P_1 = 4, P_2 = 7$
-

- For $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$:
 - $\det(A) = 0$
 - Inverse does **not** exist (singular matrix)



Summary

- A matrix is a way to organize data into rows and columns.
- Matrix operations help simplify and solve systems of economic equations
- Inversion is crucial for solving systems when direct substitution isn't feasible
- Input-output analysis is a powerful economic application
- Linear algebra is a foundation of modern data modeling and optimization



3. Home work #2



Homework #2

- **Due Date:** June 11, 2026, before the start of class.
- **Submission Format:** Submit your solutions as a single PDF file via the Cyber Campus.
- **Instructions:**
 - Clearly show all steps and calculations.
 - Include explanations for your answers where applicable.
 - Ensure your submission is neat and well-organized.
 - Bring any questions to the office hours or email me.
- Work on your Home Assignment #2 (Jacques, 10th edition, Chapters 5-9):
 - Chapter 5.1: Exercise 5.1, Problem 5 (p. 411)
 - Chapter 5.2: Exercise 5.2, Problem 7 (p. 427)
 - Chapter 5.3: Exercise 5.3, Practice Problem inside the chapter (p. 432 only)
 - Chapter 5.4: Exercise 5.4, Problem 5 (p. 456)
 - Chapter 5.5: Exercise 5.5, Problems 6 and 7 (p. 467)
 - Chapter 5.6: Exercise 5.6, Problem 4 (p. 479)
 - Chapter 6.1: Exercise 6.1, Problem 3 (p. 506)
 - Chapter 6.2: Exercise 6.2, Problem 6 (p. 521)
 - Chapter 7.1: Exercise 7.1, Problem 5 (p. 553)
 - Chapter 7.2: Exercise 7.2, Problem 6 (p. 572)
 - Chapter 7.3: Exercise 7.3, Problem 5 (p. 584)
 - Chapter 8.1: Exercise 8.1, Problem 6 (p. 615)
 - Chapter 8.2: Exercise 8.2, Problem 3 (p. 626)
 - Chapter 9.1: Exercise 9.1, Problem 4 (p. 655)
 - Chapter 9.2: Exercise 9.2, Problem 3 (p. 670)

Good luck!

Next Class

- (June 2) Cramer's Rule (7.3)





Any QUESTIONS?

Thank you for your attention and participation!